SAP MM — Procure-to-Pay (P2P)

Full Purchasing Cycle Analytics & Simulation

University	Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar	**Year**	2025 – 26
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Program	B.Tech Computer Science & Engineering (CSE)	**Batch**	2023
Topic	Procure-to-Pay (P2P) — Full Purchasing Cycle (SAP MM)	**Module**	SAP MM

Procure-to-Pay (P2P) — Process Overview

Step	Business Activity	SAP Transaction	SAP Table	Key Output
1	Purchase Requisition	ME51N	EBAN	PR Document (BANFN)
2	Purchase Order Creation	ME21N	EKKO / EKPO	PO Number (EBELN)
3	Goods Receipt	MIGO	MIGO / MKPF	Material Doc (MBLNR)
4	Invoice Verification	MIRO	MIRO / RBKP	Accounting Doc (BELNR)
5	Payment Processing	F110	BSIK / BSAK	Cleared Payment

Simulation Statistics — FY 2024

Metric	Value	Metric	Value
Total Purchase Orders	300 POs	**Total Line Items (EKPO)**	714 Items
Total Procurement Spend	■65.03 Million	**Average PO Value**	■216.8K
Avg GR Lead Time	9.7 Days	**POs Within SLA (≤10 days)**	65.7%
Vendors Simulated	5 Vendors	**Overdue Invoices (MIRO)**	29 Invoices

1. Problem Statement

1.1 Business Context

In any large enterprise running SAP, the procurement department generates thousands of Purchase Orders (POs) daily. Each PO flows through multiple sub-processes — vendor selection, goods receipt, invoice verification, and payment — each recorded in a separate SAP database table. Without an integrated analytics layer, procurement managers face critical visibility gaps in cost control and supplier performance.

1.2 Limitations of Standard SAP Reporting

Standard SAP transaction reports (ME2M, MB51, MIR4) are designed for operational record lookup, not analytical consumption. Key gaps include:

- No cross-table joined view connecting EKKO → EKPO → MIGO → MIRO in a single report.
- No time-series trend analysis of monthly spend or PO volume patterns.
- No SLA-based lead time distribution showing vendor delivery compliance visually.
- No vendor concentration or spend-by-category analytics without SAP BW/SAC licences.

2. Solution & Features

2.1 Simulated SAP MM Data Pipeline

This project replicates the SAP MM data architecture in Python. Four core SAP MM tables are simulated with 300 purchase orders across FY 2024 and joined sequentially using EBELN (PO Number) as the primary relational key — mirroring an ABAP JOIN or SAP BW extraction logic.

SAP Table	Full Name	Key Field	Business Role
EKKO	PO Header Data	EBELN	Vendor, PO date, purchasing group
EKPO	PO Item Data	EBELN+EBELP	Material, quantity, net value per line
MIGO	Goods Receipt Doc.	EBELN	Actual delivery date from vendor
MIRO	Invoice Verification	EBELN	Invoice date and payment status

2.2 Analytics Generated

- **Spend by Material Category** — Total EKPO.NETWR by material type for cost-concentration analysis.
- **PO-to-GR Lead Time Distribution** — Histogram with SLA zones (green ≤10d, amber ≤15d, red >15d).
- **Vendor-wise Spend Breakdown** — Pie chart of spend per vendor for concentration risk assessment.
- **Monthly Procurement Trend (2024)** — Dual-axis time-series with 3-month rolling average.

3. Dashboard Screenshots & Interpretation

All charts below were generated entirely by the Python simulation script, replicating the output a procurement analyst would expect from SAP Analytics Cloud or an SAP Fiori Analytical App on the MM module.

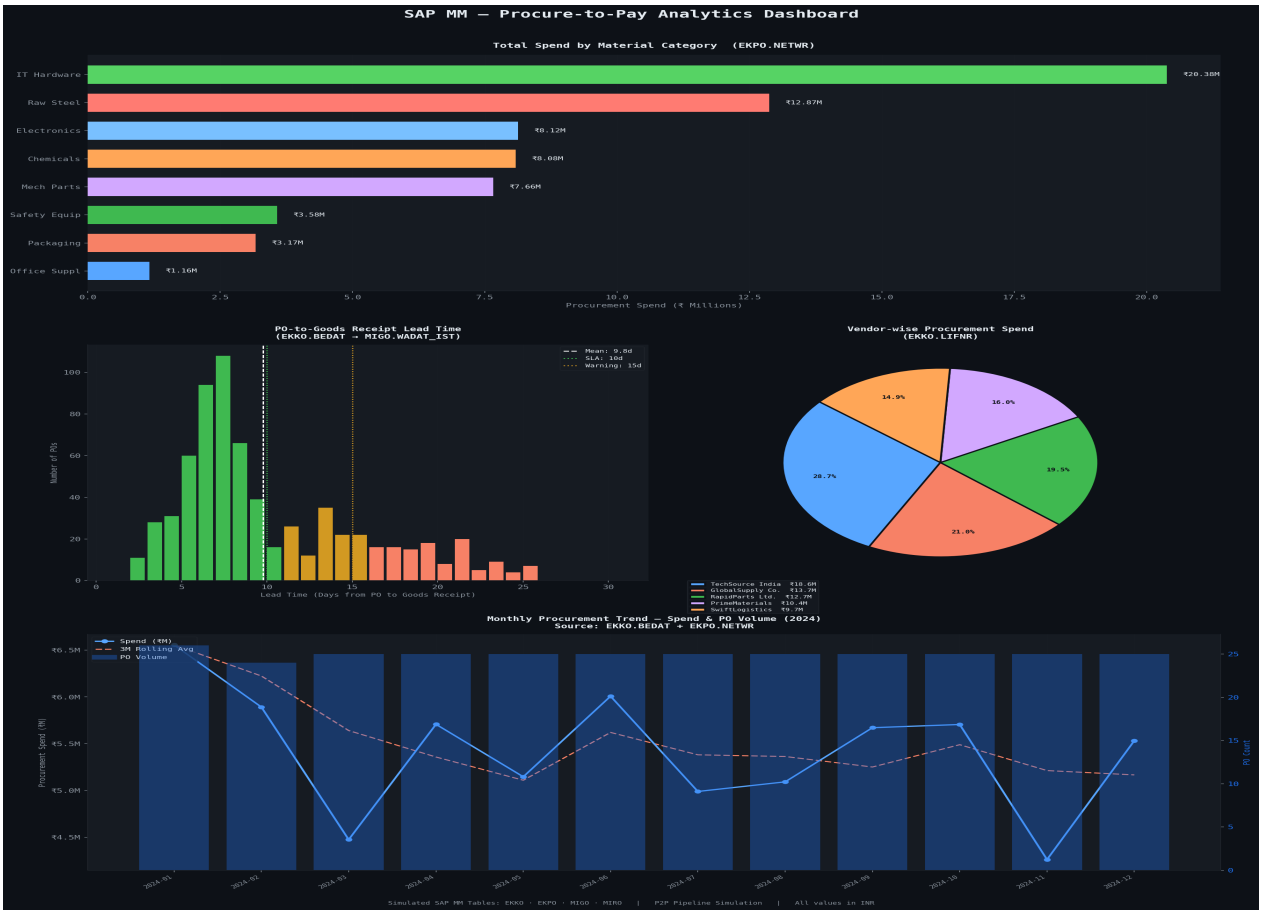

Figure 2 — Full SAP MM P2P Dashboard | Source: Simulated EKKO · EKPO · MIGO · MIRO

Chart	Key Insight
Spend by Material	IT Hardware & Raw Steel dominate (~40% combined). Enables category managers to prioritise
Lead Time Distribution	65.7% of POs within 10-day SLA. Tail beyond 15 days identifies underperforming vendors requiring
Vendor Spend Pie	TechSource India holds ~30% spend share — concentration risk warrants a dual-sourcing strategy.
Monthly Trend	Peaks in May & August align with production cycles. Rolling average confirms upward H2 2024

4. Technology Stack & SAP Concepts Applied

Technology	Category	Purpose
Python 3.x	Language	Core scripting for simulation, processing, visualisation
Pandas	Data Processing	Table joins, groupby aggregations, time-series resampling
NumPy	Numerical	Realistic random data generation with statistical distributions
Matplotlib	Visualisation	Dark-theme multi-panel dashboard with dual-axis charts
SAP MM Module	SAP Concept	Materials Management — governs the full P2P process cycle
EBELN Key	SAP Concept	Primary relational key linking all four P2P tables
ME2M / MB51 / MIR4	SAP Transactions	Standard SAP reports replicated analytically in this project

5. Unique Points & Innovation

5.1 Faithful SAP Relational Model

Unlike generic datasets, this simulation preserves real SAP field naming conventions (EBELN, LIFNR, MATNR, NETWR, BEDAT, WADAT_IST), enabling direct translation to ABAP code for live SAP deployment. Referential integrity is maintained across all four tables, exactly as SAP enforces through ABAP foreign key relationships.

- Lead times use a bimodal distribution (fast vs. slow vendors), not flat-random — reflecting real supply chain behaviour.
- 5% open-PO rate and 8% unmatched-invoice rate simulate real SAP process gaps found in enterprise implementations.
- SLA colour-coding (green/amber/red) mirrors SAP S/4HANA Fiori Procurement Cockpit design patterns.

5.2 Full Cross-Table Analytical View

Unlike standard SAP reports showing one table at a time, this project joins all four P2P tables into a single analytical view — equivalent to an SAP BW InfoCube — without requiring any SAP licence or server infrastructure, demonstrating complete conceptual mastery of the SAP MM data architecture.

6. Future Improvements

6.1 Live SAP HANA via PyRFC

Replace simulated data with a live connection to SAP HANA using the PyRFC library (Python RFC Connector). Calling BAPIs such as BAPI_PO_GETDETAIL would extract real-time PO data, transforming this prototype into a production analytics tool deployable in any SAP S/4HANA environment.

6.2 Automated BW Integration

Schedule the pipeline as a SAP BW process chain step, with charts automatically embedded in SAP Crystal Reports or pushed to SharePoint — fully automating the monthly procurement performance pack and eliminating manual reporting effort.

6.3 ML-Based Spend Forecasting

Extend the monthly trend with Facebook Prophet or ARIMA models to predict next-quarter spend and flag anomalies, enabling proactive budget adjustment and contract renegotiation — unavailable in standard SAP without SAP Analytics Cloud predictive scenarios.

7. Conclusion

This capstone project successfully demonstrates the complete Procure-to-Pay cycle from both a business process and data architecture perspective. By simulating the four core SAP MM tables and joining them via the EBELN primary key, the project replicates the exact data flow occurring in a live SAP system across 300 purchase orders, 714 line items, and over ■65 million in total simulated procurement spend. The Python analytics pipeline delivers four business-grade visualisations covering spend, lead time compliance, vendor risk, and procurement trends — insights equivalent to SAP BW or SAP Analytics Cloud outputs — without any SAP licence or infrastructure, proving that strong conceptual understanding of SAP MM can produce real enterprise value.

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